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Identifying the pollution sources of River Yamuna through field observations and participatory approach

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River Yamuna, flowing through the National Capital Territory (NCT) of Delhi, is considered as one of the worst polluted river stretches in the world. The river is primarily polluted through untreated sewage discharge from various point and non-point pollution sources that find their way through storm drains and join the river Yamuna. One of the major requirements to address urban water quality is to identify the potential sources of contamination in a large drainage basin, which is challenging as the basin area is so big that it is difficult to pinpoint the source of pollution. Invariably, as the pollutants move into the drains under gravity, it is possible to identify the drains and their corresponding catchment areas that have higher pollution load. Thus, the study aims to find the sources of pollution in a larger river basin by integrating field observation, GIS approach, and qualitative interviews. Water samples from 43 monitoring locations are collected in the various storm drains of the Barapullah drainage basin, in various primary, secondary, and tertiary drains, and analyzed the water quality parameters in the Delhi Jal Board (DJB) water quality laboratory. The locality-wise spatial contribution of pollution is mapped in the context of the Barapullah basin. The study results found excessive BOD values ranging from 37 to 430 mg/L, and six major pollution hotspots were identified. The qualitative interview conducted in the six major pollution hotspots regarding the source of pollution and results indicated that the discharge of domestic sewage from planned colonies, unauthorized colonies, and JJCs, as well as market waste disposal, are the potential reasons for pollution in the six major pollution hotspots. This pilot-level study provides a useful and effective approach to investigating and identifying pollution sources, mitigating the pollution at the source, and ultimately reducing the pollution level at river Yamuna.

Keywords. Yamuna; river health; environment pollution; water quality; NCT Delhi.

1. Introduction

Water is an essential element for human survival and a key input in balancing the ecological system of the global environment. A country's social and economic progression depends on the availability of freshwater within the country (Sarker *et al.* 2022). This important resource has been exploited extensively in recent decades, resulting in depletion (Qian *et al.* 2021), pollution (Das *et al.* 2022), and

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destruction of freshwater sources like rivers, lakes, tanks, and groundwater (Chotpantarat and Thamrongsrisakul 2021). One of the major global environmental challenges is to prevent the contamination of river water. Poor river water quality threatens the aquatic environment and creates fatal consequences for human beings. With rapid economic development, increasing population, and urbanization, river water is getting polluted from various point and nonpoint pollution sources such as soil erosion, disposal of untreated effluent, surface runoff, chemicals from agricultural and industrial activities, etc. (Aggarwal *et al.* 2022).

In urban areas, where separate sewerage and stormwater infrastructure have been established, improper or illegal non-stormwater discharge into the stormwater drainage system may be significantly responsible for polluting the receiving waters, especially during dry weather (Xu *et al.* 2014; Yin *et al.* 2019). During wet-weather days, along with surface runoff from precipitation that drains from the city, inappropriate discharge of wastewater exaggerates the overflow of pollution (Yin *et al.* 2019). In urban cities, untreated sewage discharge in storm drains is one of the major components of river water pollution, especially in developing countries. Despite imposing strict laws and setting up various standards, untreated sewage is directly discharged into the natural drainage system, causing pollution and worsening of its quality (Jadeja *et al.* 2022; Sinha 2023; Tariq and Mushtaq 2023). Restoring the natural drainage system is a major challenge to the government and environmentalists.

Therefore, one of the major steps to address urban stormwater quality is to identify the potential sources of contamination in a drainage basin. Identifying and terminating the pollution at sources can reduce large-scale river pollution. Identification of the potential source of contamination can be achieved through the assessment of the water quality of different stormwater drains in a drainage basin (Yan *et al.* 2015) so that the high-risk areas or the pollution hotspots get identified and the government can take effective actions by checking the pollution at sources.

Many big cities in India are located close to the rivers and invariably discharge huge amounts of wastewater into these rivers resulting in the degradation of the quality of the river water (Srinivas *et al.* 2020; Ghosh *et al.* 2021). River Yamuna, one of the most sacred rivers in India, flows through the National Capital Territory (NCT) of Delhi and is considered one of the dirtiest river stretches in the world (Patel *et al.* 2020). The Yamuna River, from its origin from the Yamunotri glacier up to the Wazirabad barrage (23 km upstream of its entry point in Delhi), has good river water quality as per CPCB standards (Mutiyar *et al.* 2018). River Yamuna enters NCT Delhi near Palla in the north and leaves NCT Delhi at 6.9 km downstream of Okhla barrage in the south, near Jaitpur. This 22 km river stretch between Wazirabad and Okhla, once considered the 'lifeline' for the capital city of Delhi, is now considered as the river 'about to die' (Misra 2010). This 2% river stretch carries 71% of its total wastewater load (Mutiyar and Mittal 2014; Govt of NCT Delhi 2020). NCT Delhi is one of the most populated cities after Tokyo and Mumbai, attracting millions yearly. Along with the rapid urbanization and population growth, many unplanned settlements with poor infrastructures are prevalent, and new ones are coming up. The major cause of the high pollution level in the NCT Delhi stretch of river Yamuna is the untreated sewage discharge from unauthorized colonies, slums, industries, punctured sewer lines, etc., which find their way to storm drains and join the river Yamuna (Verma *et al.* 2022; CPCB 2023). This problem is further exacerbated by the lack of sewerage facilities in many places, the insufficient capacities of sewage treatment plants (STPs) against the actual sewage generation, improper location of STPs, and inconsistencies in the functionalities to the optimum level (due to technical and maintenance problems) (Kumar *et al.* 2019; CPCB 2021). Currently, Yamuna receives 218 million gallons of domestic sewage daily from NCT Delhi through various stormwater drains (NCTD 2021). To reduce the pollution level of Yamuna, it is important to identify the source of pollution so that appropriate action is taken to plug the same. Since the sewage is flowing into stormwater drains under gravity and each drain has its contributing area, it is possible to identify as which area is bringing in more pollution by conducting a water quality assessment.

Several studies have been done in India on water quality assessment and identifying high-risk zones. Girija *et al.* (2007) assessed the water quality of the Bharalu river, a tributary of the Brahmaputra river in Guwahati, and identified the location where the contamination exceeds the permissible limits. In the context of NCT Delhi, several studies have been carried out on water quality assessment of the Yamuna river (Sharma *et al.* 2008; Mandal *et al.* 2010; Sharma and Kansal 2011; Katyal *et al.* 2012; Kaur *et al.* 2013; Rout 2017; Bhat *et al.* 2018; Patel

et al. 2020). 60% of Delhi's wastewater enters the Yamuna through the Najafgarh drain. The drain is unlined for 51 km before reaching the Yamuna. Recent measures limit groundwater withdrawal and artificial recharge. The region's groundwater is contaminated, making it unsafe for drinking (Shekhar and Sarkar 2013). Vaid *et al.* (2022) studied the water quality status of the Najafgarh drain and its various secondary drains in NCT Delhi and identified comparatively more polluted secondary drains.

However, these studies are related to finding the potential sources of pollution of the Yamuna River in NCT Delhi, which has not been of much help since the identified area is so big that it is difficult to pinpoint the source. Also, in the context of NCT Delhi, the water quality assessment has been reported in the main drainage outlet (CPCB 2023), but the in-depth assessment of the secondary drains and further upstream has not yet been studied. Water quality assessment at the basin level requires not only an investigation of water pollution at the main drainage but also the recognition of the pollution level at the secondary and tertiary drains so that the identification of polluted risky regions is possible. Therefore, if the pollution sources are to be plugged, then locality-wise spatial contribution of pollution is mapped so that appropriate action is taken to plug the same. Therefore, identifying pollution hotspots by mapping high-contributing localities and their catchments using a geographic information system (GIS) will benefit decision-makers by prioritizing more focus on the high-risk areas and adapting proper mitigation actions to clear storm drains of pollution and, ultimately, river Yamuna. So, this is the first of its kind of study in the context of NCT Delhi to analyze the wastewater quality in the pilot study basin of Barapullah along with its associated primary, secondary, and tertiary drains; and identify the most pollution-contributing localities and explore the probable reason for the pollution by integrating water quality assessment, GIS mapping approach, and qualitative interview with the local communities. This will benefit decision-makers to focus on the high-risk areas and derive proper mitigation actions to improve the condition of the storm drains and, thereby, ultimately, river Yamuna. So, the main objectives of the study are: (i) to identify the areas/sub-catchments that significantly contribute to the pollution and (ii) to investigate the probable source of pollution in the identified hotspots by engaging with the stakeholders.

2. Study area, materials and methods

2.1 *Profile of the study area*

NCT Delhi has three major drainage basins, i.e., Najafgarh, Barapullah, and Trans-Yamuna River. The Barapullah River basin has been chosen as the pilot-level study area among the three basins. Barapullah basin is located in the southern part of the NCT Delhi (figure 1). Barapullah basin is surrounded by the Yamuna River in the east, Haryana state in the south, and Najafgarh basin in the north and west, having a catchment area of 376.27 km^2. As per the 2011 Census, the total population in this basin is approximately 3.39 million, covering mainly five districts, i.e., South Delhi, South-west Delhi, North Delhi, Central Delhi, and New Delhi. The catchment area is mainly urbanized, except for a few areas in the southern part. Among various stormwater drains, the Barapullah drain is one of the major drains that flows eastwards to join the Yamuna. It discharges 33 million gallons of domestic sewage daily to Yamuna (Bhaduri 2016). It receives much of its wastewater from high-density residential areas, many of which are still not connected to sewerage systems, such as Sangam Vihar, Mehrauli, and Sarai Kale Khan.

2.2 *Data used*

A Shuttle Radar Topography Mission Digital Elevation Model (SRTM DEM) with 30-m resolution for the Barapullah basin is used to delineate the watershed and the stream network for the Barapullah basin (https://earthexplorer.usgs.gov/). This is validated using the natural drainage map from the Department of Irrigation & Flood Control (I&FC), Govt. of Delhi (https://ifc.delhi.gov.in/). After verifying it with Google Earth imagery, monitoring locations are identified. A spatial map of the Jhugri Jhogri Colonies (JJCs, informal settlements/slums) is obtained from the Delhi Urban Shelter Improvement Board (DUSIB). The locality map of NCT Delhi is obtained from the publicly available GitHub repository (https://github.com/).

2.3 *Sampling sites and strategies*

In order to identify the highest pollution-contributing areas, water quality monitoring is performed in different stormwater drains in the Barapullah basin in a systematic manner, starting with the first-order

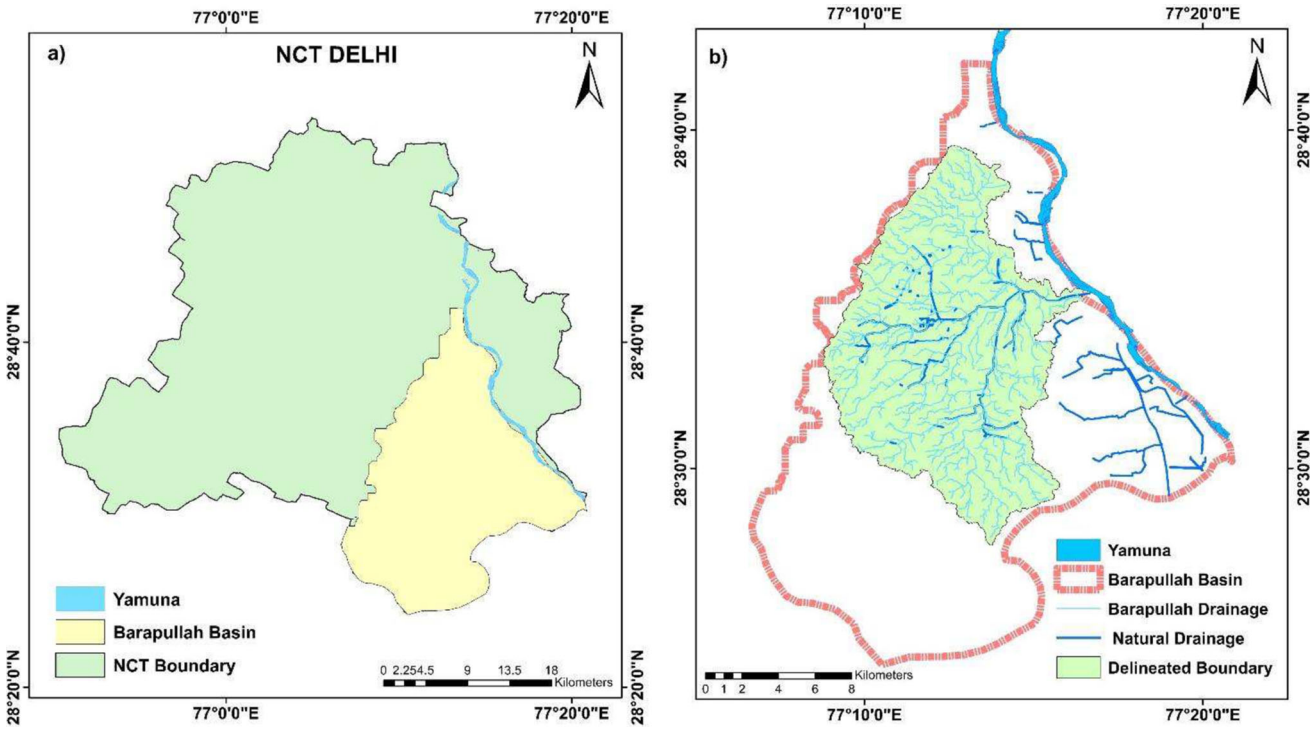

Figure 1. **(a)** Map of NCT showing Barapullah basin and **(b)** Barapullah sub-basin with the natural/storm network.

drains and moving downstream to identify those sub-areas of the Barapullah basin that are contributing comparatively higher levels of pollution. Monitoring locations are identified based on the drainage network. In the case of the confluence of two drains, samples are taken in both the drains before and after their confluence to identify which drain is bringing in more pollution from its contributing area. It is also important to mention that a large part of the stormwater drains in the Barapullah basin are covered. The field team was not able to take the samples in those drains, which are covered and physically inaccessible. Because of this constraint, the study is limited only to a part of the entire drainage system of the Barapullah basin. Within the Barapullah basin, 43 monitoring locations (S1–S43) are selected based on the physical accessibility of the drains. The geographic coordinates and details of all the 43 locations are mentioned in table 1.

In order to avoid the dilution effect of rainfall, sampling is done in all 43 monitoring locations in the non-monsoon period from January to May 2022. Except for a few sites, the samples are mostly collected in the peak hours in the morning (9 am–12 pm). Peak hours are chosen for sampling, as domestic sewage flow is highest in most cities in the morning and evening hours (Penn *et al.* 2012). Samples from the monitoring locations are collected and stored in polypropylene bottles. The bottles are thoroughly cleaned and rinsed with distilled water before collection. The collected samples are labelled properly. The coordinates of each sampling site are noted using handheld GPS. The samples were brought immediately to the water quality laboratory of the Delhi Jal Board at the Office of the Chief Water Analyst III, Delhi Jal Board, in Okhla Tank, Mathura Road, New Delhi-110020. Delhi Jal Board is the government agency responsible for producing and distributing drinking water and collecting, treating, and disposing sewage throughout NCT Delhi. In surface water environment monitoring, parameters including total organic carbon (TOC), chemical oxygen demand (COD), and biochemical oxygen demand (BOD) are crucial indicators for gauging the level of water pollution (Bourgeois *et al.* 2015). Among them, the BOD is one of the most often used and significant indicators of organic contaminants in water, making it a valuable tool for assessing water quality (Chee 2013; Lee *et al.* 2016; Wang *et al.* 2020). Water quality will deteriorate due to increased organic matter, raising the BOD concentration and decreasing dissolved oxygen (Wang *et al.* 2020). BOD is the amount of oxygen consumed by bacteria and other micro-organisms (Li *et al.* 2019) as

Table 1. *Location details of 43 monitoring locations in Barapullah drainage basin, NCT Delhi, India.*

Loc	Lat.	Long.	Description
S1	28.55305	77.1648	CGHS Colony Vasant Vihar
S2	28.56851	77.1896	Block A2, Safdarjung Enclave
S3	28.56942	77.1979	Block A-25, Nauroji Nagar
S4	28.54168	77.2264	Block E, Saomi Nagar South Ext.
S5	28.53212	77.2249	PSRI Hospital, Malviya Nagar
S6	28.5533	77.2299	Archana Arcd, Greater Kailash
S7	28.57526	77.2186	Rishi Nagar Extin
S8	28.57959	77.2236	Sewa Nagar
S9	28.57976	77.2365	Block C, Lajpat Nagar
S10	28.58694	77.2477	Hanuman Mandir Nizamuddin
S11	28.54591	77.1871	Masala Mix (Iit Delhi)
S12	28.5445	77.1959	Kailash Hostal(Iit Delhi_)
S13	28.52784	77.2256	Govt. Quarter Pushp Vihar
S14	28.52577	77.2163	Saket Dist. Centre
S15	28.51824	77.2255	Ambedkar Marg Pushp Vihar
S16	28.52087	77.2248	Sec-5 Pushp Vihar
S17	28.55992	77.2311	Andrewsganj Delhi
S18	28.56495	77.232	Mahatma Gandhi Andrewsganj
S19	28.58622	77.2451	Jangpura Bhogal
S20	28.58683	77.2437	Jangpura Pant Nagar
S21	28.49345	77.226	Devli Sangam Vihar
S22	28.48757	77.2299	L2-Block, Devli Sangam Vihar
S23	28.53117	77.22376	Sec-6 Saket (PSRI)
S24	28.521599	77.225693	Sector 5 Pushp Vihar
S25	28.515051	77.23494	Ambedkar Nagar I
S26	28.512055	77.243812	Block D Sangam Vihar
S27	28.578846	77.236281	Sewa Nagar Railcolony
S28	28.579175	77.245748	Jal Vihar Lajpat Nagar
S29	28.553089	77.185567	Sector 3 RK Puram
S30	28.560986	77.184212	Adrash Colony RK Puram
S31	28.570197	77.182979	Mudirka Marg RK Puram
S32	28.56769	77.177174	Ekta Vihar RK Puram
S33	28.571963	77.172592	Somnath Marg RK Puram
S34	28.562217	77.167989	Block A Vasant Vihar
S35	28.560094	77.167748	Basant Lok Vasant Vihar
S36	28.562742	77.201656	Arjun Nagar, Ansari Nagar
S37	28.56054	77.209201	Yusuf Sarai Gautam Nagar
S38	28.573563	77.215282	East Kidwai Nagar
S39	28.520212	77.237032	Ambedkar Nagar II
S40	28.535949	77.222999	Sheikh Sarai Village
S41	28.603319	77.196586	Teen Murti Marg
S42	28.590256	77.240646	Bus Depot Nizamuddin
S43	28.584052	77.26609	Sarai Kale Khan

they act upon the organic matter under aerobic conditions at a specified temperature (Abhyankar *et al.* 2020). Vigiak *et al.* (2019) have assessed solely the BOD as the spatial analysis of the water quality trends. As the storm drains of NCT Delhi mainly carry sewage, BOD can be a good indicator for identifying the pollution hotspots in the Barapullah basin. In our study, we analyzed the samples for Biological Oxygen Demand (BOD), one of the widely used parameters in measuring the quality of sewage (Hur *et al.* 2010; Akpor and Muchie 2011; Zare Abyaneh 2014; Abhyankar *et al.* 2020). BOD is determined for all the samples, as per IS 3025, Part-44. Sampling is done again in the second phase to check the reliability of the result.

2.4 *GIS approach and community interview*

In order to identify the localities contributing significantly to the pollution, we have overlaid the locality map of NCT Delhi over the sub-catchments of each monitoring location so that the contributing catchments inside the highly polluted localities can be explored extensively to pinpoint the pollution sources. In order to evaluate any contribution of sewage discharge from JJCs in the storm drains, spatial maps of JJCs are also overlaid. All the layers are overlaid in a GIS platform.

Once the highly polluted localities or the pollution hotspots were identified, the research team conducted a qualitative interview to understand the probable sources of pollution in those hotspots. An open-ended questionnaire has been prepared for the local communities living in those hotspot areas to identify the probable source of pollution in those highly polluted sub-catchments and localities. The questionnaire consists of a few open-ended questions regarding understanding the community perceptions on the major contributors of pollution in that drain, such as unauthorized colonies or planned colonies, any local markets, or any industries. Each respondent is first introduced to the purpose of the study, followed by questions like 'Do you think the drain is polluted in your locality?' and 'What is the major reason for the pollution of the drain in your locality?' A total of 18 in-depth interviews are conducted with the local residents in six identified 'major pollution hotspots' in the Barapullah basin in the month of July 2022. Each of the interviews lasted for about 20 minutes. Respondents are selected through purposive sampling in each of the major pollution hotspots. The finding of the qualitative interview is discussed briefly in section 3: Results and discussion. The probable sources of pollution in each hotspot, as investigated through the qualitative interview, are mentioned in table 2.

3. Results and discussion

3.1 *Overview of the water quality assessment*

The water quality of 43 sampling sites is assessed based on the BOD levels. The BOD values are found to be excessively high, ranging from 37 to 430 mg/L, as against the CPCB standard of a maximum of 3 mg/L for most of the sampling sites in the Barapullah basin. The variation of BOD values in these 43 sites is illustrated in figures 2 and 3(a). The highest BOD concentration was observed in monitoring location S36 (Arjun Nagar, Ansari Nagar), which is 430 mg/L, followed by S26 (Block-D Sangam Vihar), S25 (Ambedkar Nagar I), and S21 (Devli Sangam Vihar), where BOD concentrations were observed at 235, 200, and 188 mg/L; respectively which is significantly high as per the international standard such as World Health Organisation (WHO) standard value of 5 mg/L of surface water quality parameters (Kankal *et al.* 2012).

Although all the monitoring locations have very high BOD levels compared to the CPCB standard

Table 2. *Summary of the probable source for pollution in the 'major pollution hotspots' found through qualitative interviews with the local communities.*

Sl. no.	Hotspots	Locations	Localities	Outcome of interview
1	Hotspot 1	S36	Safdarjung Enclave Ansari Nagar West	Discharge of domestic sewage from the planned colonies to the undercovered drain by pumping from SPS of Delhi Jal Board.
2	Hotspot 2	S26 S25 S21 S22	Sangam Vihar	Discharge of domestic sewage from unauthorized colonies and highly dense slums of Sangam Vihar via open drains.
3	Hotspot 3	S31 S32 S29	Ramakrishna Puram	JJCs adjacent to the drain directly discharge their sewage into the drains. Waste disposal (vegetables & meat) from the Indra Market.
4	Hotspot 4	S5 S40	Sheikh Sarai	Discharge of domestic sewage from JJCs/slums.
5	Hotspot 5	S15 S16	Pushpa Vihar	Contaminated water from highly dense Sangam Vihar from upstream causes pollution.
6	Hotspot 6	S8	Sewa Nagar	Discharge of domestic sewage from JJCs.

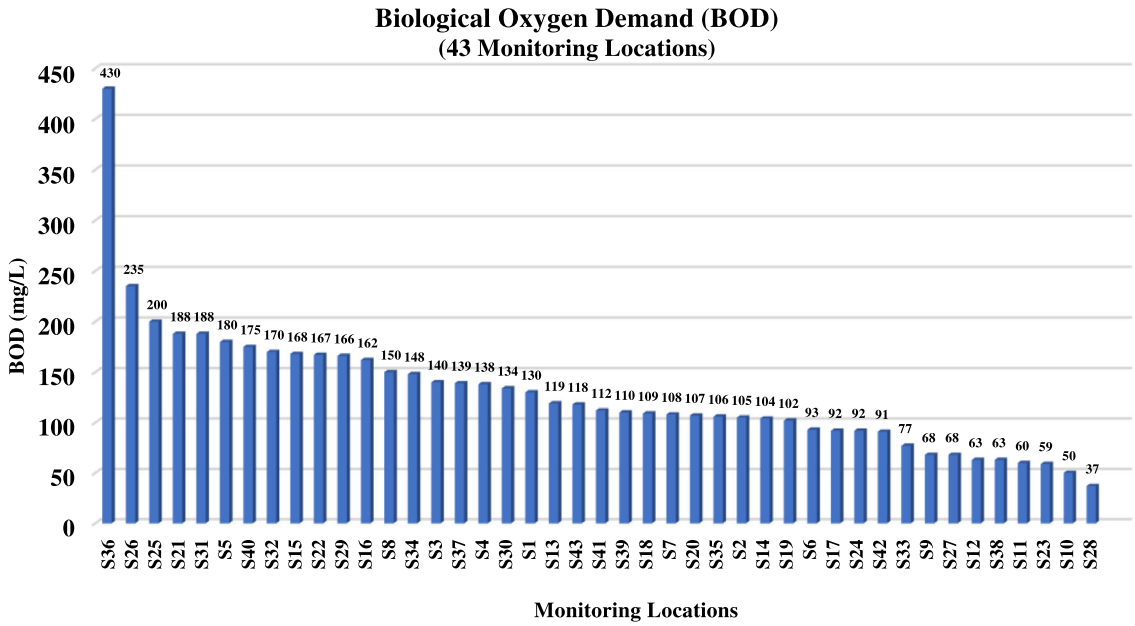

Figure 2. BOD values of 43 monitoring locations in the Barapullah basin (source: Water Quality Laboratory, Delhi Jal Board, Okhla Tank).

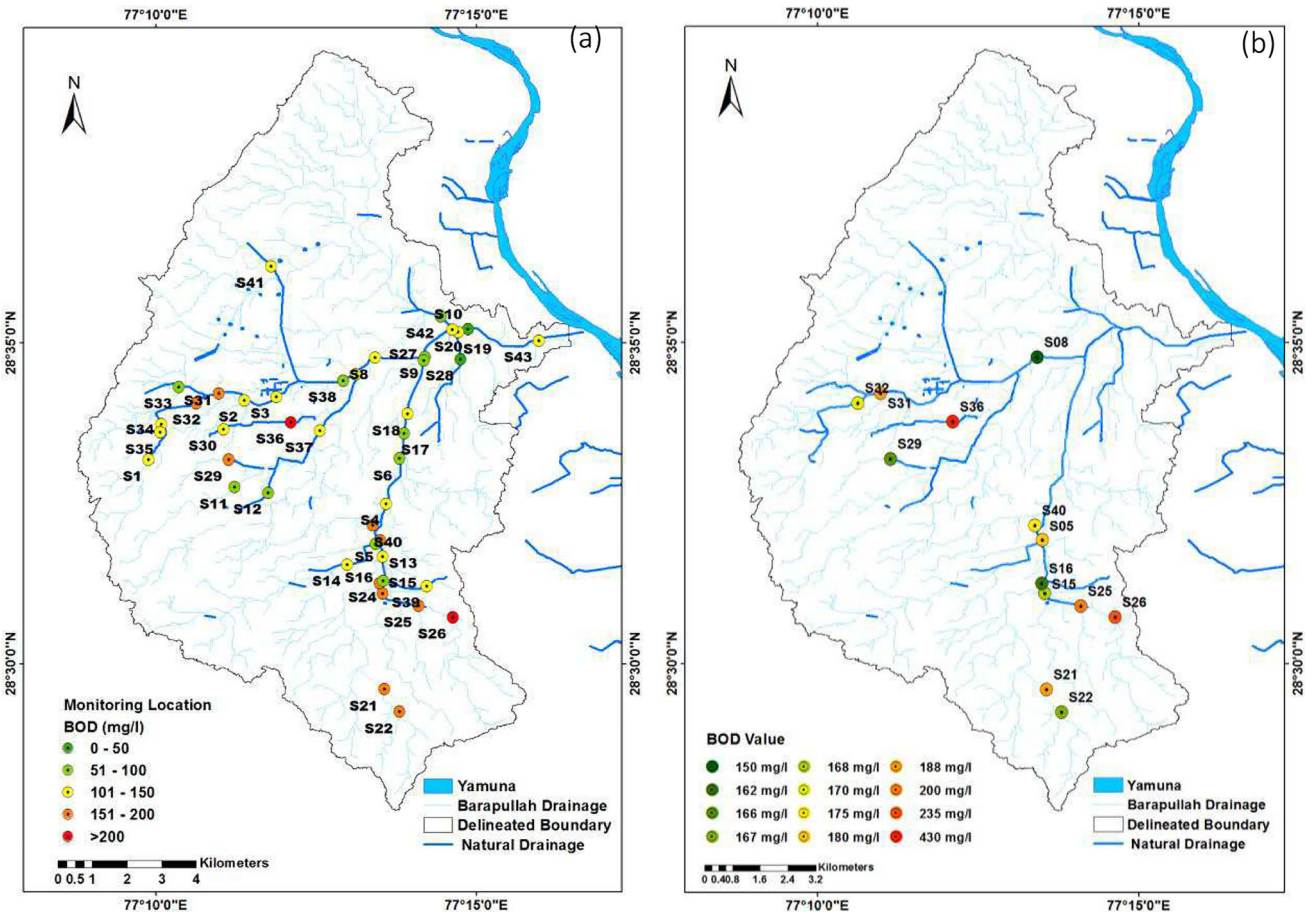

Figure 3. Spatial map of monitoring locations and BOD values **(a)** all 43 locations and **(b)** 13 hotspot locations.

of 3 mg/L, we have considered a BOD value of 150 mg/L as the threshold for identifying 'highly polluted locations' in this study. Our main concern is identifying the sources of pollution, primarily due to raw sewage directly discharged into various storm drains in NCT Delhi (Verma *et al.* 2022). Therefore, we have used the lower limit of BOD in raw sewage, 150 mg/L, as specified in the Streamkeeper's Field Guide (Murdoch *et al.* 1996). Thirteen monitoring locations have a BOD value higher than 150 mg/L and are shown in figure 3(b). To ensure the reliability of the laboratory test results, sampling was done again in the second phase for those 13 monitoring locations. It was found that all those 13 monitoring locations recorded BOD values of more than 150 mg/L in the second phase as well.

3.2 *Overview of 'major pollution hotspots' in the Barapullah basin*

At the basin level, to identify the sub-areas of the localities that are responsible for contributing to higher levels of pollution, the GIS layers of the locality map of NCT Delhi and their associated sub-catchment layers are overlaid (figure 4). Based on the 13 critical locations, 6 hotspot sub-catchment areas ('major pollution hotspots') were identified as the high pollution contributing localities or 'major pollution hotspots'. Following this, we conducted qualitative interviews with the local communities residing in the subareas to identify the probable cause of pollution. The outcome of the interviews is mentioned in table 2. The highest BOD concentration is observed in Arjun Nagar Nallah at Safdarjung Enclave and Ansari Nagar West, i.e., 430 mg/L, followed by Sangam Vihar, where BOD concentration is observed in three different monitoring locations, i.e., 235 mg/L, 200 mg/L, and 188 mg/L, respectively. The details of the localities with the highest BOD records are described below.

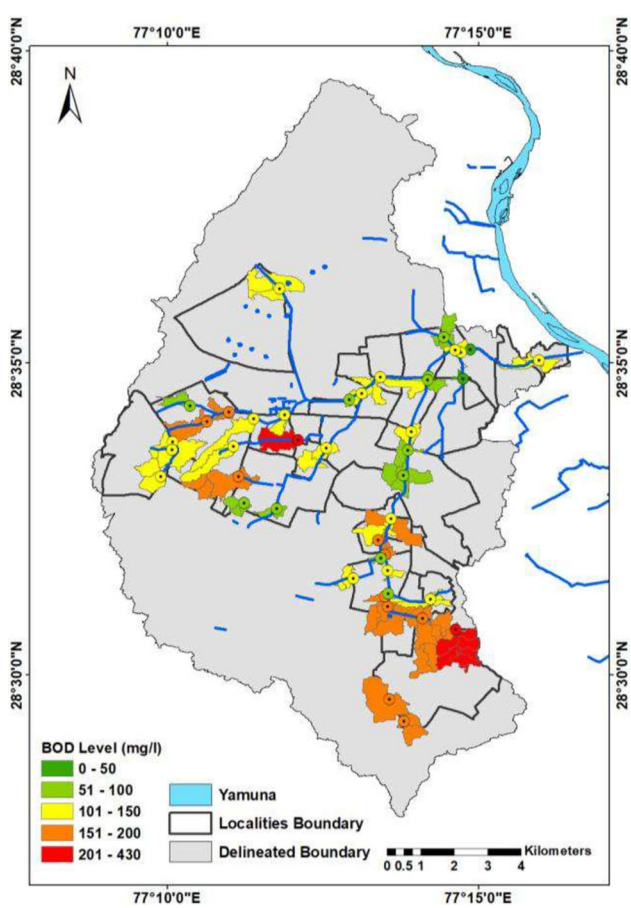

Figure 4. Spatial contribution of pollution of localities and its associated sub-catchments.

3.2.1 *Hotspot 1: Ansari Nagar West and Safdarjung Enclave*

The highest BOD concentration, i.e., 430 mg/L, is observed in the Ansari Nagar West locality. The catchment or contributing area lies in the central part of the locality of Safdarjung Enclave and a portion of the southern parts of Ansari Nagar West (figure 5a). The sampling site is located in Arjun Nagar Nallah, Arjun Nagar, Ansari Nagar (S36). Based on the field observation, the upstream area of the sampling site or catchment area, primarily consists of residential buildings or planned colonies, where the drain is mostly undercovered.

Rigorous interviews with the residents of the Safdarjung Enclave and Ansari Nagar West reveal a few probable causes of significant pollution in the Arjun Nagar Nallah. As per the residents of the two localities, the residential buildings on both sides of the undercovered Nallah directly discharge their domestic sewage to the undercovered drain. A 36-year-old resident living in Safdarjung Enclave mentioned, '*The main reason for the pollution is because of the sewage generated from the flats and apartments built on both sides of the covered part of the Nallah. Those residential buildings directly discharge their domestic sewage into the covered Nallah, that is why this drain is so polluted*'. For those houses, this nallah works as a sewer system. Along with that, a few residents also conveyed that there is one Sewage Pumping Station (SPS) of Delhi Jal Board at Arjun Nagar (28.562522°N, 77.1983°E), which directly pumps

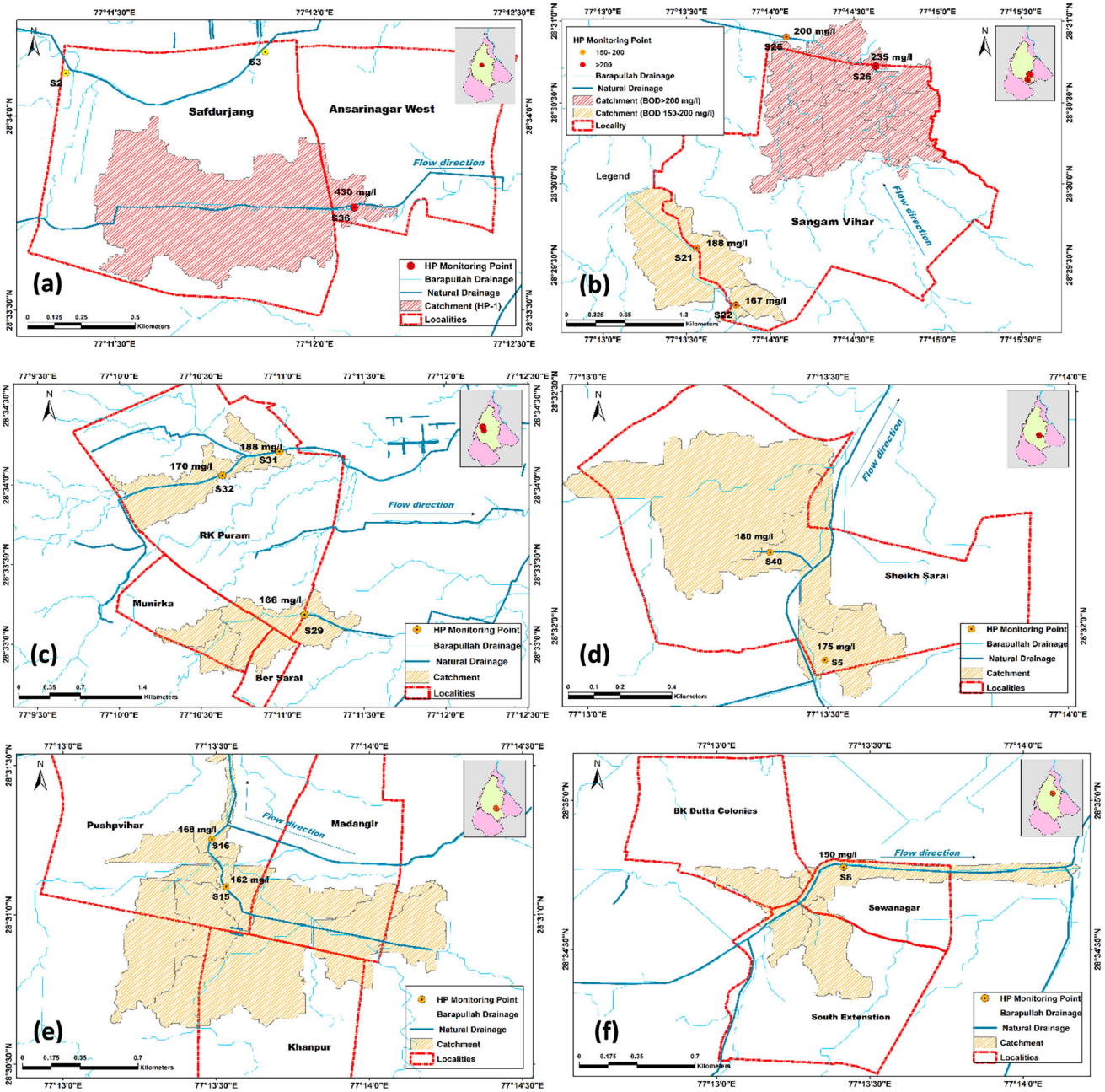

Figure 5. Details of sub-catchment and localities area of **(a)** Hotspot 1, **(b)** Hotspot 2, **(c)** Hotspot 3, **(d)** Hotspot 4, **(e)** Hotspot 5, and **(f)** Hotspot 6.

and discharges the sewage of the entire locality upstream of the monitoring location into the undercovered Arjun Nagar Nallah, instead of pumping to the sewerage network to connect to the STP. Again, a resident in Arjun Nagar mentioned that "*Every day the sewage pumping station in Arjun Nagar, where all the household's wastewater is collected, directly discharge the household's wastewater into the nallah, which is undercovered and not visible from outside*".

3.2.2 *Hotspot 2: Sangam Vihar*

According to the ranking of monitoring locations based on BOD values, Sangam Vihar has the second, third, and fourth-highest BOD concentrations. The BOD values of each monitoring location in Sangam Vihar are as follows: Block D Sangam Vihar (S26) has the second highest BOD level, 235 mg/L; Dr. Ambedkar Nagar, Dakhsinpuri (S25) has the third highest BOD level, 200 mg/L; and Devli Sangam Vihar (S25) has the fourth highest

BOD level, 188 mg/L (S21). In southwest Sangam Vihar, another location named L2 Block-Devli Sangam Vihar (S22) records a BOD concentration of 167 mg/L, ranked 10th out of 13 'major pollution hotspots'.

Sangam Vihar is a high-density unauthorized colony located in the southern part of the Barapullah basin at South Delhi with a total area of roughly 5 km^2. The catchment areas for the first two monitoring locations (S26 and S25) are in the northern Sangam Vihar. Other two-monitoring locations (S21 and S22) are located in the southern part of the Sangam Vihar (figure 5b). As per the field observation, the catchment areas of the monitoring locations consist of majorly informal settlements, which lack proper civic amenities.

The in-depth interviews with the local communities reveal that households directly discharge domestic sewage in the small drains (1–2 ft. wide and 2–2.5 ft. deep) located beside the local street in Sangam Vihar that ultimately join to the larger drains. According to one resident, the slope of Sangam Vihar divides the area into two parts. The northern and central parts of the Sangam Vihar slope towards the main road in the north. At the same time, the southern and eastern part slopes towards the southwest. The main reason Sangam Vihar is one of the major hotspots is because of the high-density informal settlements discharging untreated domestic sewage to the major drains in the northern and southwestern parts of Sangam Vihar. A resident in Sangam Vihar mentioned "*Our house is not connected to underground sewer lines; in fact, no houses here are connected with underground sewer-line. All the domestic wastes go through small street-side open drain that ultimately join the main drain. During the monsoon it becomes a big problem as all the streets are flooded with wastewater*". Another respondent from Sangam Vihar mentioned "*Sangam Vihar is so huge and densely populated that lots of wastewater get generated every day that passes through the street side narrow drains. A huge part of the wastewater flows north-direction, and rest of it flows towards the south and ultimately joins the nallah*".

3.2.3 *Hotspot 3: Ramakrishna Puram*

Among the 13 'major pollution hotspots', the 5th highest BOD level is recorded in Mudirka Marg (S31) at Ramakrishna Puram, i.e., 188 mg/L. The sampling site is located in the eastern part of the Ramakrishna Puram. The contributing catchments are located mainly in the central and eastern parts of Ramakrishna Puram (figure 5c). Field-observation and overlaying of the GIS layer of the JJ Colonies reveal the existence of the informal settlements like slums/Jhugri Jhogri Colonies (JJCs) and unauthorized colonies in the catchment area, that discharge their domestic sewage to the stormwater drain and make it one of the highest polluted areas within the Barapullah basin.

We have selected another monitoring location for water quality sampling at 650 meters (approx.) upstream of Mudirka Marg (S31), in the same drain (figure 5c), at Ekta Vihar, RK Puram (S32). This location records BOD value of 170 mg/L and is ranked 8th among the 13 'major pollution hotspots'.

During the in-depth interviews, local communities revealed that the drain goes through the middle of the JJCs in Mudirka Marg (S31), that have no civic amenities. The JJCs discharge their domestic sewage directly to the adjacent open drain with connected pipelines from their households (figure 6a). A woman living in the informal settlement of Mudrika Marg mentioned, "*The drain here is primarily polluted due to the discharge of domestic sewage from the slums adjacent to the drain. Every household has its own pipeline for discharging its wastewater into the drains. We have no other options than to live with the drain. It will be good for us if the drain will be covered*". A few residents also revealed that one of the primary reasons for the high pollution in that location is the waste disposal (vegetables, fish, and meat) from the local market named Indra Market (28.565704°N, 77.167567°E), which is located upstream of the two monitoring locations in Ramakrishna Puram. For example, a resident in the slum of Mudirka Marg mentioned "*The major reason for the pollution in this locality is because of the Indra-market where all the market thrash like meat and vegetables wastes are thrown to the drain; that creates a problem here. Sometimes it is unbearable.*" During the field visit, we also found vegetables and meat markets adjacent to the drain and evidence of market waste disposal on the drain (figure 6b).

In the extreme southern part of the Ramakrishna Puram, another monitoring location i.e., Sector-3, RK Puram (S29), has been selected for water quality sampling, where the BOD concentration was found to be 166 mg/L. Among the 13 'major pollution hotspots' (with BOD>150 mg/L) this

Figure 6. Source of pollution and interview. **(a)** Discharge of domestic sewage directly to the adjacent open drain with connected pipelines from the households in Mudirka Marg (S31), **(b)** Waste disposal directly to the adjacent open drain from the local meat market named Indra Market (S31 and S32), **(c)** In-depth interview with the local community beside the Arjun Nagar Nallah in Arjun Nagar, Ansari Nagar.

location ranked 11th. The overlaying of the catchment, locality map, and JJCs layers reveals that the water mainly comes from the informal colonies or JJCs in the eastern part of Munirka. The northeastern part of Ber Sarai and the southern part of RK Puram also lie in the catchment area (figure 5c).

3.2.4 *Hotspot 4: Sheikh Sarai*

As per the ranking of the monitoring locations based on BOD values, the 6th and 7th highest BOD concentrations are observed in Sheikh Sarai. The BOD results of each of the monitoring locations at Sheikh Sarai are as follows: the 6th highest BOD level, i.e., 180 mg/L, is recorded at PSRI Hospital (S5), and the 7th highest BOD level, i.e., 175 mg/L, is recorded at Sheikh Sarai village (S40). The contributing catchments are located mainly in Sheikh Sarai's southern and eastern-central parts (figure 5d). Field observation and overlaying of the GIS layer of the JJ Colonies reveals the existence of the informal settlements upstream of the monitoring location (S40), which can be a significant contributor to the high levels of pollution.

During the interview, the local community revealed that untreated domestic sewage from the JJC named Jagadamba Camp is adding pollution to the stormwater drain at Sheikh Sarai village (S40). A respondent living closer to the monitoring point in Sheikh Sarai mentioned, "*All the polluted water comes from the Jhuggis in Jagadamba Camp*". In another location near PSRI Hospital, communities are mostly unaware of the reason for the high level of pollution. Based on the field observation in the contributing catchment areas of

the monitoring location near PSRI Hospital, we found that the catchment area mainly consists of greenery, the main road, and a hospital (PSRI Hospital).

3.2.5 *Hotspot 5: Pushpa Vihar*

As per the ranking of the monitoring locations based on BOD values, the 9th and 12th highest BOD concentrations are observed in Pushpa Vihar's locality. The BOD results of each of the monitoring locations at Pushpa Vihar are as follows: the 9th highest BOD level, i.e., 167 mg/L, is recorded at Ambedkar Marg, Pushpa Vihar (S15), and the 12th highest BOD level, i.e., 162 mg/L, is recorded at Sec-5, Pushpa Vihar (S16). S16 is located 300 meters upstream of S15. The major contributing catchments are located in the southern part of Khanpur and Pushpa Vihar, and the northern part of Madangir and Sangam Vihar (figure 5e).

The in-depth interview with the community reveals the high BOD level in that location is because the drain brings highly polluted sewage water from the high-density informal colonies of Sangam Vihar, located upstream of the monitoring location.

3.2.6 *Hotspot 6: Sewa Nagar*

Among the 13 'major pollution hotspots', the 13th highest BOD level is recorded in Sewa Nagar (S8), i.e., 150 mg/L. The sampling site is located in the northern part of Sewa Nagar. The contributing catchments are located mainly in the southeastern part of BK Dutt Colony, the western part of Sewa Nagar, and a few northwestern parts of South-Extension part 1 (figure 5f). Field investigations and overlaying of the GIS layer of the JJ Colonies reveal the existence of informal settlements like slums/Jhugri Jhogri Colonies (JJCs) upstream of the sampling point that discharges their domestic sewage to the stormwater drain.

4. Conclusions

The NCT Delhi stretch of the Yamuna River has become one of the world's most polluted river stretches. Despite numerous actions taken for a long time to curb the pollution rates of Yamuna, the pollution level is still significantly higher than the CPCB standard of maximum BOD of 3 mg/L. Domestic effluents are a significant contributor, mainly adding pollutants to the river Yamuna. Previous studies have assessed the spatial and temporal variation in water quality at River Yamuna and its major drain outlets but failed to identify the sources of pollution. In a drainage basin, it is difficult for decision-makers to pinpoint the pollution source as it is huge in size. Terminating the pollution at sources can be one of the effective solutions to stop the large-scale river pollution of Yamuna. It is important to identify and locate pollution sources to stop pollution at its source. In our study, we have taken a novel approach to solve the issue by taking the Barapullah basin of NCT Delhi as a case study and assessing the water quality at the secondary and tertiary drains to identify the most polluted risky regions. By integrating GIS, we have mapped the high-contributing localities and their catchments that can help the decision-makers by prioritizing more focus on the high-risk areas and adapting proper mitigation actions to clear storm drains of pollution and, ultimately, river Yamuna. Based on the BOD results, this study attempts to find the pollution sources and hotspots in a pilot-level river basin, i.e., the Barapullah river basin. Within the pilot-level study area, the study found that the major pollution hotspots are located in Safdarjung Enclave, Ansari Nagar West, Sangam Vihar, Ramakrishna Puram, Sheikh Sarai, Pushpa Vihar, and Sewa Nagar, where BOD values have been recorded more than 150 mg/L, in two phases. The results also revealed a few potential pollution causes in those major pollution hotspots. In high-density locations like Sangam Vihar, which has no sewerage facilities, there is huge pollution to the storm drains travelling downstream. The situation is also similar for slums in Ekta Vihar and Mudirka Marg, which are a source of contamination of storm drains. Installing decentralized sewage treatment plants at major pollution hotspots can decrease the pollution loads in stormwater drains and eventually help reduce large-scale pollution in the Yamuna River. The study may help policymakers establish and identify potential locations where decentralized sewage treatment plants can be installed within the Barapullah basin. Once the decentralized sewage treatment plants in the pollution hotspots are installed, gradually, it can help improve the environmental health of the stormwater drains at the Barapullah basin and the river Yamuna. It will also help to strengthen the previous restoration plans like the 'Yamuna Action

Plan', 'Mailey Se Nirmal Yamuna Revitalization Plan 2017', 'Interceptor Sewer Project', and several 'NGT orders'. Our approach can be adapted as a methodology to investigate and identify pollution hotspots in any river basin of the world. Extending the study to the whole of NCT Delhi will give a clearer picture of the other pollution sources in Delhi and ultimately help the policy-makers to reduce the pollution of Yamuna to a large extent by taking proper intervention strategies like installing decentralized sewage treatment plants in those identified pollution hotspots.

However, the study has several limitations regarding data and sampling strategies. The study has only used one-time sampling for the 43 monitoring locations (twice for only 13 locations). Repetitive sampling in all the monitoring locations and capturing temporal variation of water quality in each location will give a clearer picture of the reliability of the results. Moreover, most of the drains in the Barapullah basin are covered, and the 43 monitoring locations are selected based on the physical accessibility of the drains. As sampling many drains was not possible due to inaccessibility, the study portrays only a partial picture of the Barapullah basin. Selecting more monitoring locations in all the drains (including undercovered drains) will give better results. Moreover, the analysis was done based only on BOD results. Detailed investigation of other water quality parameters (such as other physiochemical, microbiological, and heavy metal concentration parameters) may give a better picture of the pollution hotspots and their potential causes. Lastly, the results of qualitative interviews with the local communities only depict the probable causes; the empirical evidence on the probable causes also needs to be identified.

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Author statement

All the authors contributed to the preparation of the manuscript. Sayantan Samui collected the primary data and prepared the initial draft. Arkadip Mallik collected data and contributed in preparing of maps and graphs. Dinesh Kumar also collected data. Sandhya Rao prepared data regarding GIS. Dhanya C T and A K Gosain gave the ideas, guided through the work, and edited and proofread the manuscript.

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